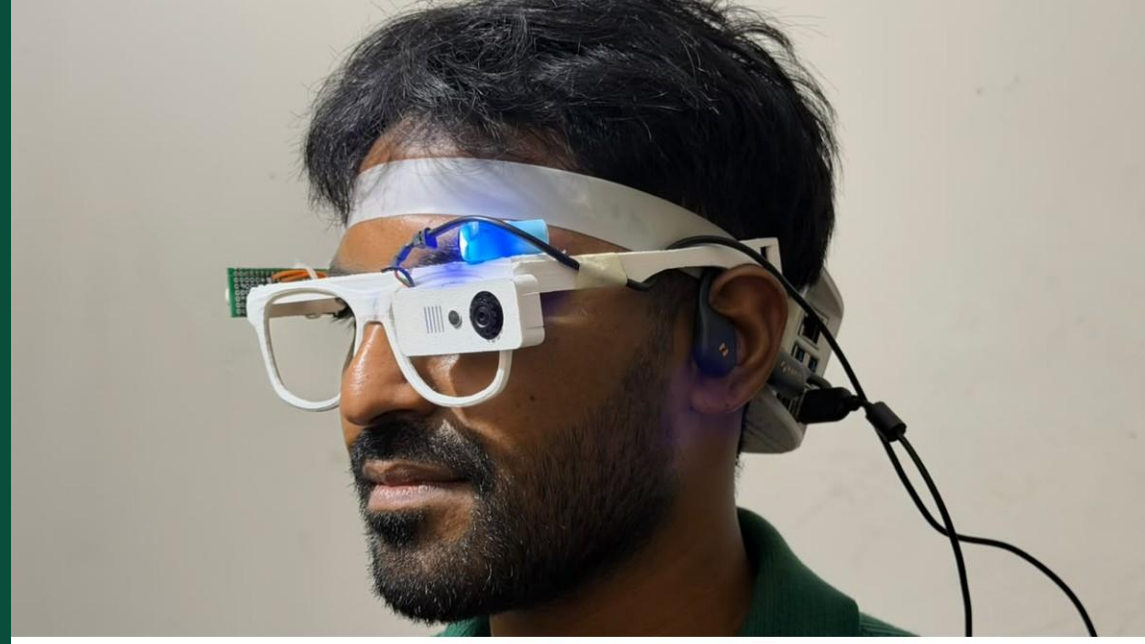
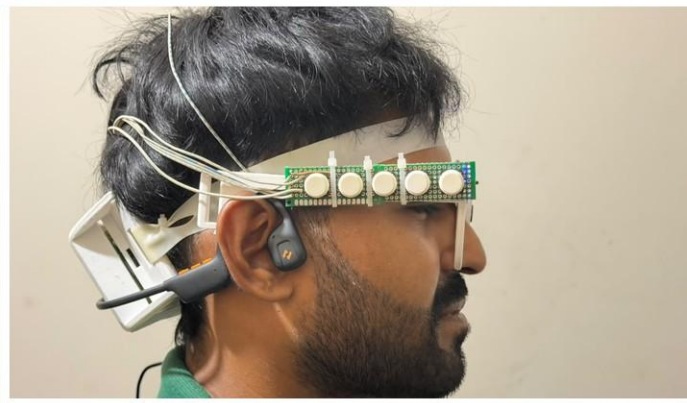
SARA · Smart Assistive Reading Aid

A Wearable Assistive Robot for Authenticating Bangladeshi Banknotes: Fusing Physical Security Features and Deep Learning for Visually Impaired Users

MD Abdun Fattah Lam (2220014) · Mukut Protim Memo (2020086)
Tanjila Hossain Tapti (2130561)
Supervisor: Mohammad Shidujaman, PhD, Assistant Professor
Department of Computer Science and Engineering, Independent University, Bangladesh









Three everyday challenges

Money

Reading

Surroundings







Illustration

Why existing solutions fall short

2.2 billion

people worldwide have a near or distance vision impairment [1]

most of them in low- and middle-income regions [1]



Limited Bangla support

OrCam MyEye does not support Bangla [2]



Internet dependence

Research glasses send pictures to a server [3, 4]

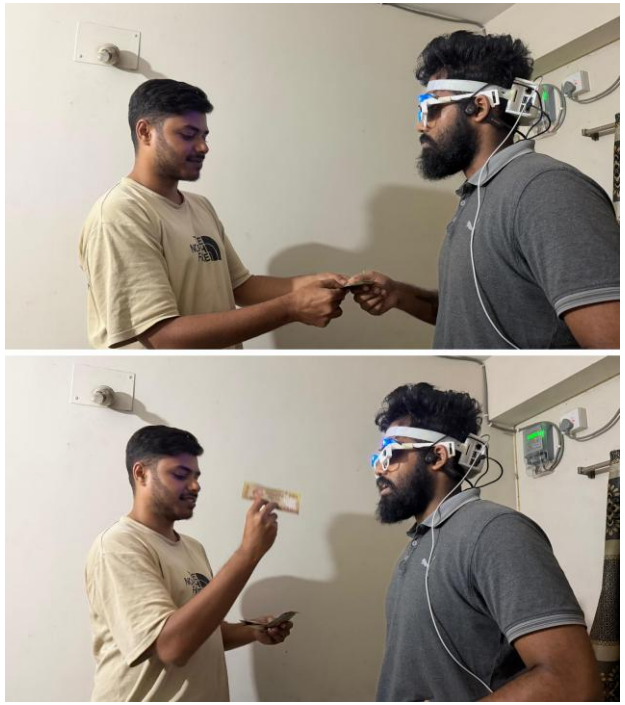


Limited Taka verification

Reviewed systems do not name Taka notes [2, 3, 4]

Sources: [1] WHO 2023; [2] OrCam MyEye; [3] Ali A. et al., IEEE Access 2021; [4] Brilli et al., arXiv 2024.

SARA in action: ask the glass, not a stranger



Our prototype in a staged shop exchange: receiving a note, holding it up to the camera

Speak
"Glass, check money"

→

Detect offline
Raspberry Pi 5, no internet

→

Hear the answer
Bangla speech + vibration

Money
Names the note and screens it with care

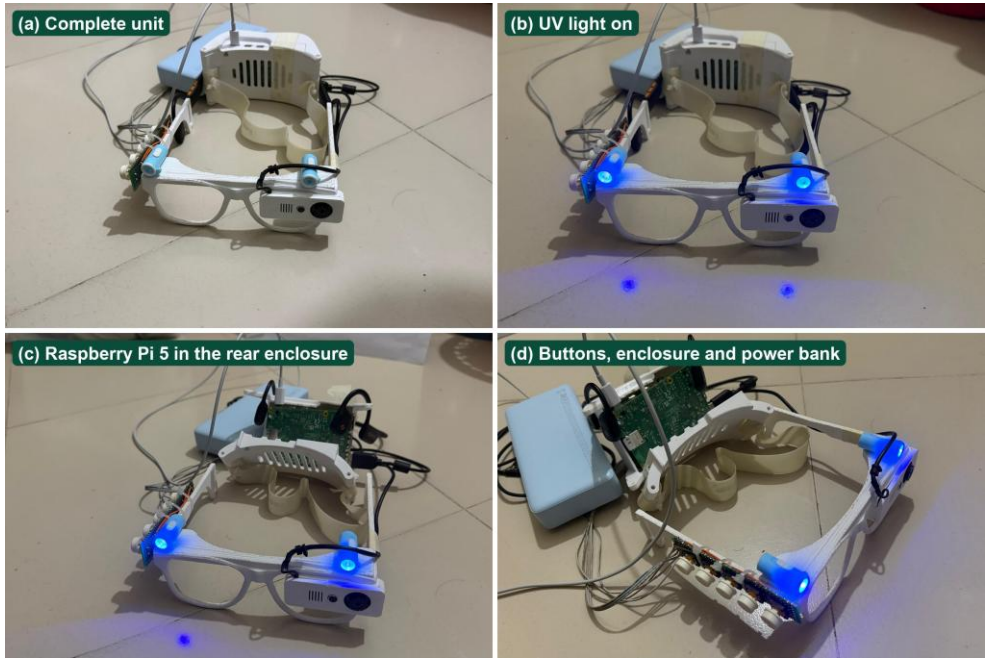
Reading
Reads Bangla and English labels aloud

Surroundings
Names people and objects ahead

পাঁচশত টাকার নোট "500 Taka note": the answer the glass speaks

Offline: text, objects, Taka, emotion. **Optional online:** scene description only.

Five modes, started by button or by voice



Prototype version 2: the complete unit, the UV light on, the Pi 5 in the rear enclosure, buttons and power bank

Text reading

Bangla and English

Objects

Named in Bangla

Taka notes

Value + safe check

Emotion

Face in front

Online description

Optional; needs internet

How the user starts a mode: five buttons on the frame, or a voice command: "Glass, read text"

Six research questions

1 System

Can it all run on a Raspberry Pi 5?

2 Synthetic data

Can unlabelled training data detect real notes?

3 Number of views

Can one view of a note be enough?

4 Unseen prints

Does the check hold on new counterfeit prints?

5 Safety

Can it never accuse a genuine note?

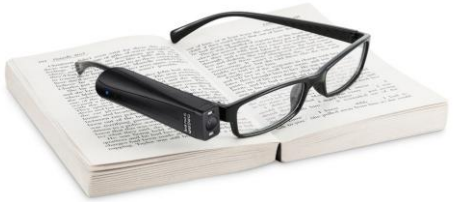
6 Bangla OCR

What matters most for reading Bangla?

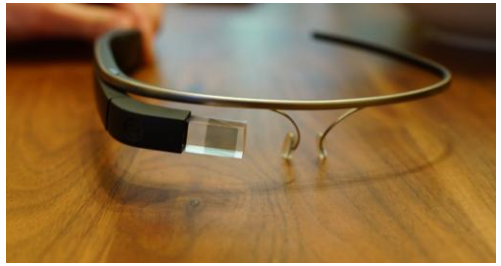
What existing devices miss

System	Reads Bangla	Knows Taka	Works offline	Wearable
OrCam MyEye [2]	No	No	Yes	Yes
Google Glass scene analysis [3]	No	No	No	Yes
AIris (2024) [4]	No	No	No	Yes
Raspberry Pi reading aids *	No	No	Partly	Partly
SARA (ours)	**Yes**	**Yes**	**Yes**	**Yes**

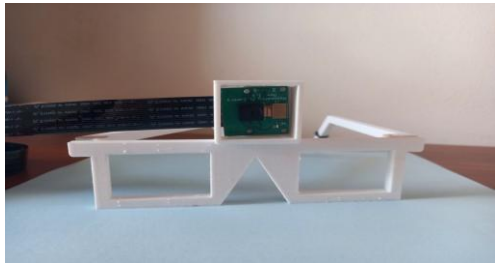
*Rows with a reference: from the cited source. * Our assessment of typical hobby builds, not a cited result.*



OrCam MyEye [2]



Google Glass analysis [3]



AIris [4]

The research gap

None of the systems we reviewed combines Bangla, Taka and offline use.

- Counterfeit tests reuse the same prints
- Few report speed on the real device

Photos: OrCam MyEye by Ylip, CC BY-SA 4.0; Google Glass by Ted Eytan, CC BY-SA 2.0 (Wikimedia Commons); AIris from [4]. Comparison from [2, 3, 4].

How the glass works

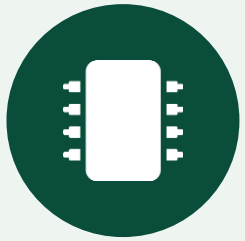


Camera, buttons, voice

Camera 640×480

5 buttons on the frame

Voice: "glass, …"



Raspberry Pi 5

Mode manager

All models on the CPU

Offline, except the online mode



AI modules

Text: CRAFT + EasyOCR [5, 6]

Objects: YOLOv8s, COCO [7, 8]

Taka: YOLOv8s + safe checks

Emotion: YuNet + EfficientNet [9]

Online: cloud model (optional)



Audio + vibration

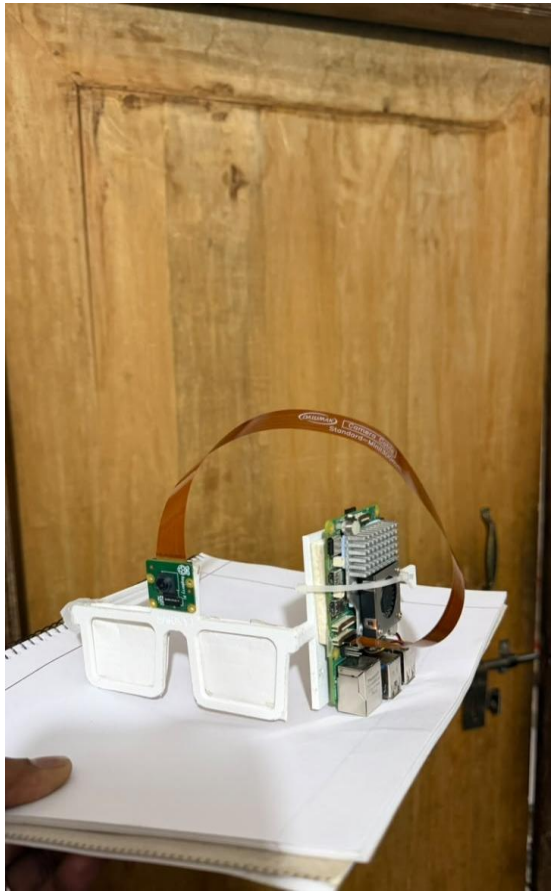
Bangla or English speech [10, 11]

Vibration for a note

Everything runs on the glass itself; only the optional online mode uses the internet.

Sources: [5] Baek et al. 2019; [6] EasyOCR; [7] Ultralytics YOLOv8; [8] COCO; [9] RAF-DB; [10] Piper; [11] eSpeak NG.

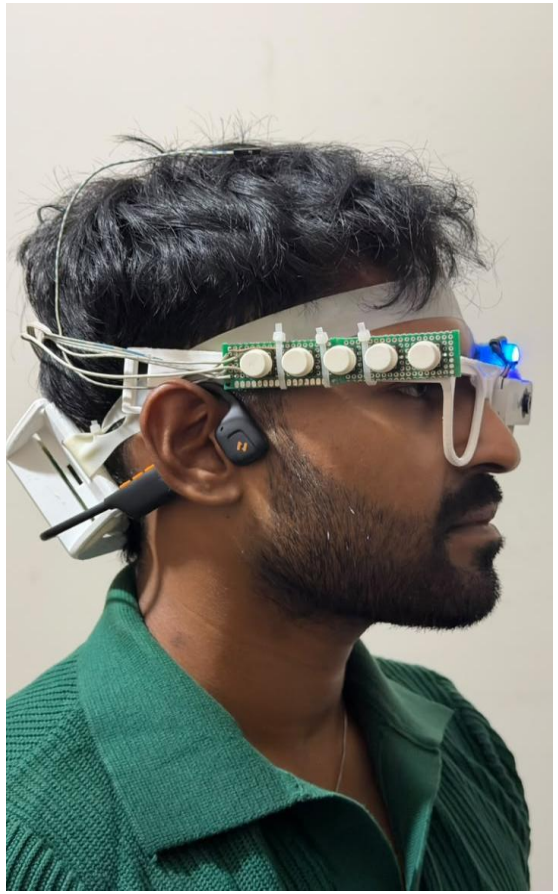
From a desk prototype to a wearable glass



Version 1
Board on one temple



Front
Camera + UV lamp



Side
5 buttons, bone-conduction earphone



Rear
Pi 5 in a ventilated case

Taka detection: 22,333 training images, no hand labels



Note photos [12] are pasted onto real scenes [8], and the program writes every label itself.

SYNTHETIC COMPOSITES

0.995

mAP@0.5 on 2,282 held-out composites

REAL PHOTOS

94.1%

right note value on 1,536 photos taken by other people (laptop)

Sources: [12] BanglaTaka (Mendeley); [8] Lin et al., COCO, ECCV 2014; detector [7] Ultralytics YOLOv8. 94.1% with the hand-held detector (next slide).

Notes held in a hand: from 1 in 8 to 1 in 2

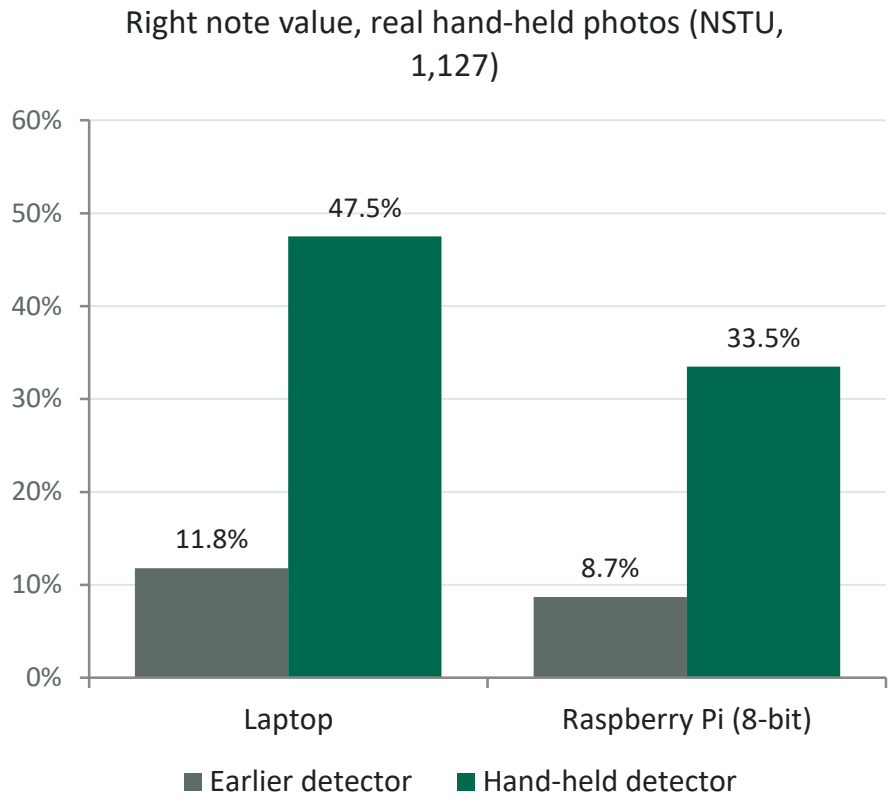
TRAINING: 6,000 SYNTHETIC CLOSE-UPS



TEST: REAL HAND-HELD PHOTOS

11.8% → 47.5%

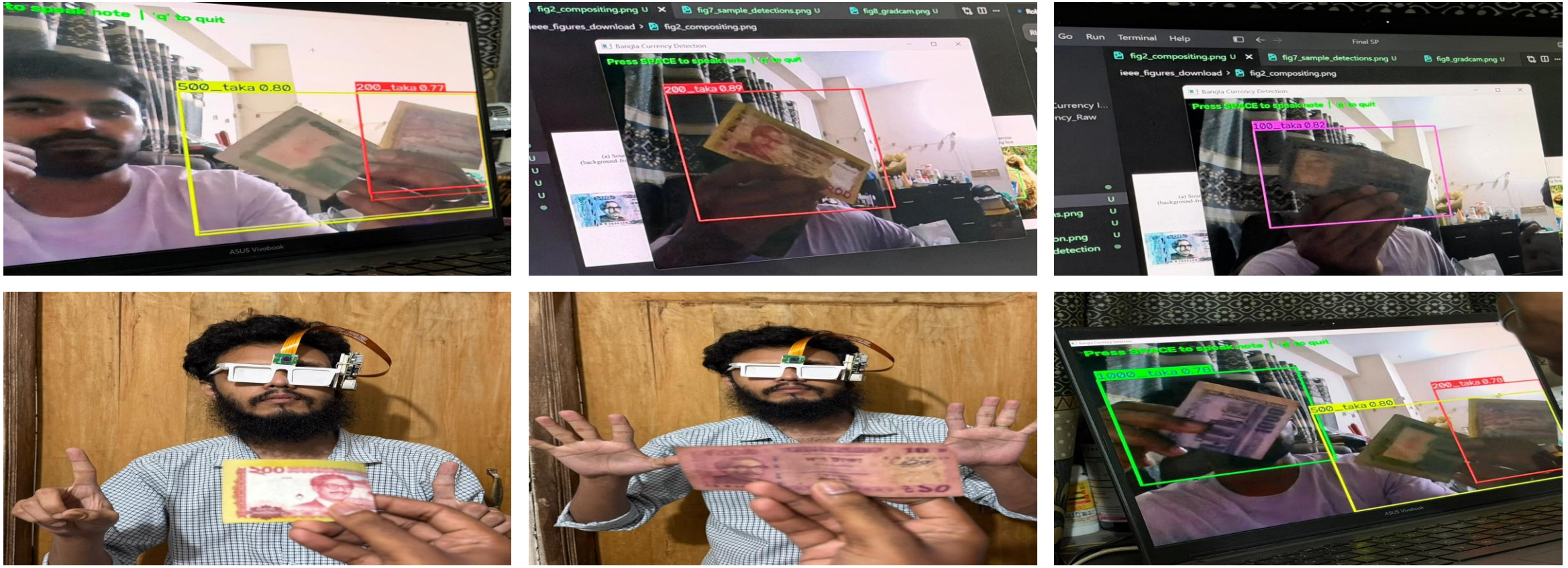
Right note value on the laptop. Trade-off: with 3 notes in view, the total is right 97.0% (was 99.2%).



Sources: test photos [13] Rafi et al., NSTU-BDTAKA, Data in Brief 2024; detector [7] YOLOv8. Our results: paper_evidence/HANDHELD_FINETUNE.md.

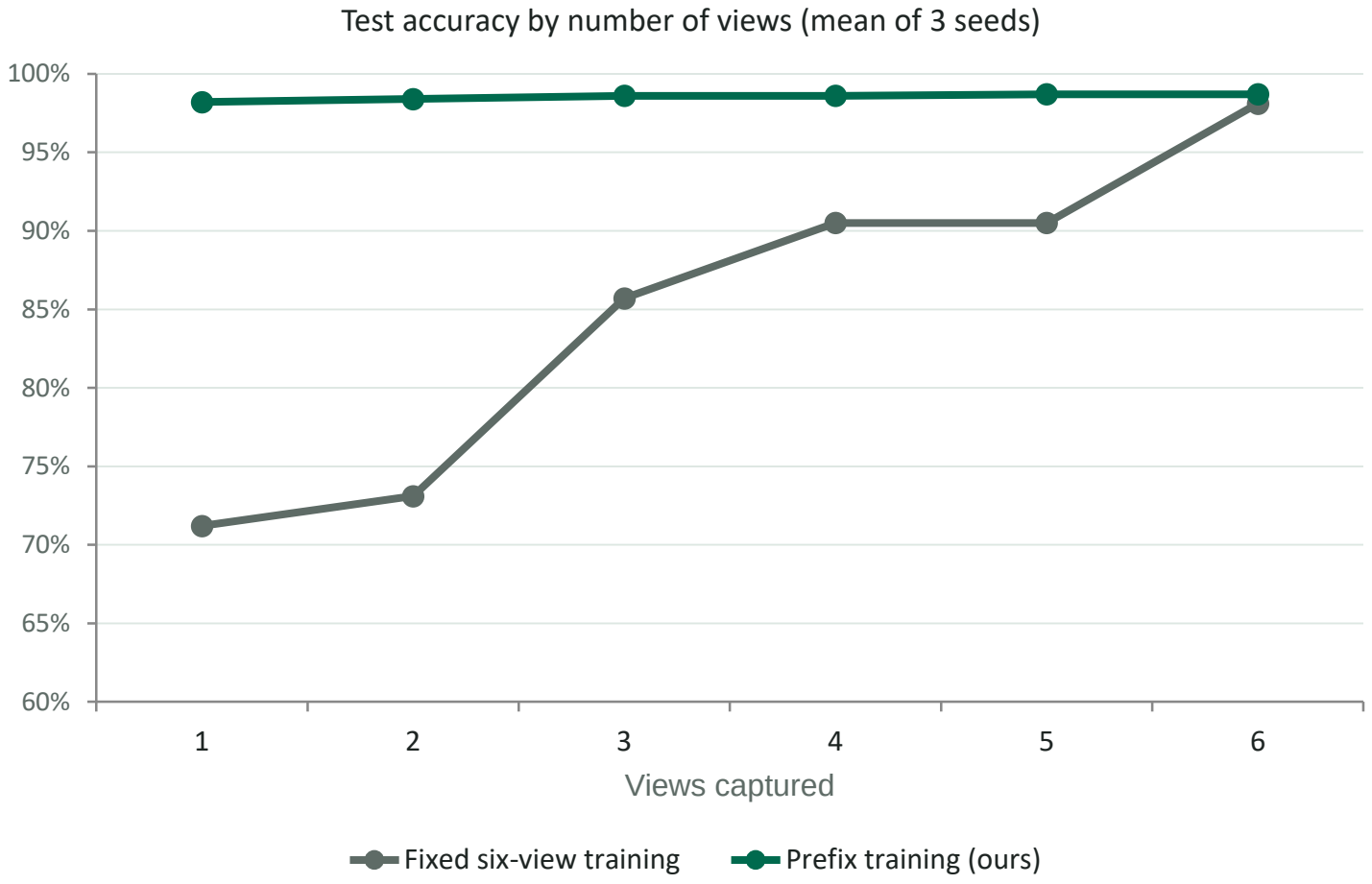
Live tests with real notes

Informal demonstrations, not a measured accuracy study



Six note values (10 to 1,000 Taka) named at 0.63–0.89 confidence, including worn and folded notes. Laptop webcam and the first glass prototype.

Counterfeit check: one view is enough with prefix training



With a single view

71.2%

rises to

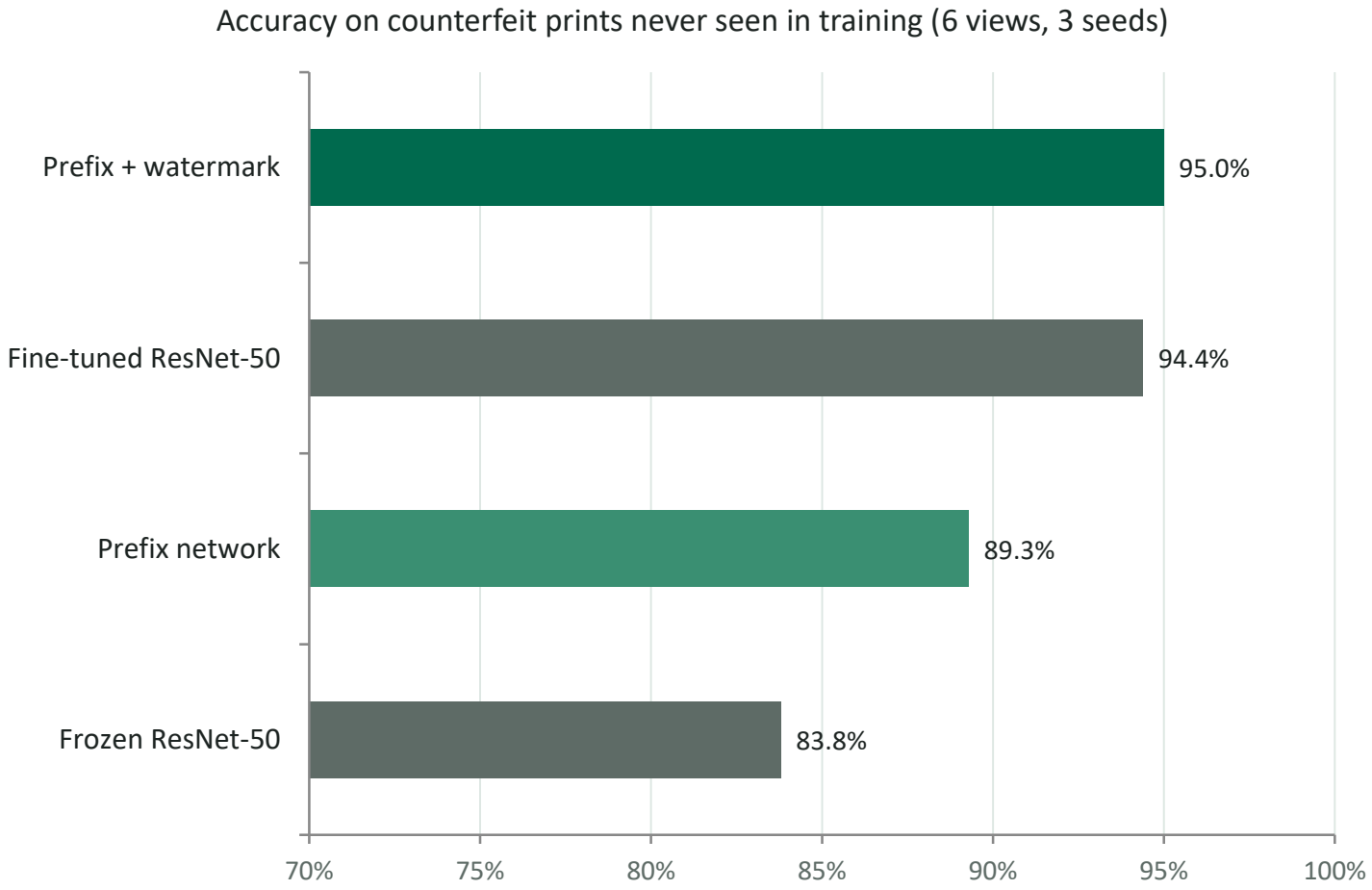
98.2%

Training on random first views: one view is enough.

Held-out JaalTaka test notes, never used for training or for any choice

Data: [14, 15] Piyas et al., JaalTaka dataset and article, 2025-2026 (1,390 notes, six views each; counterfeits seized by the Rapid Action Battalion).

Unseen counterfeit prints: a stricter test



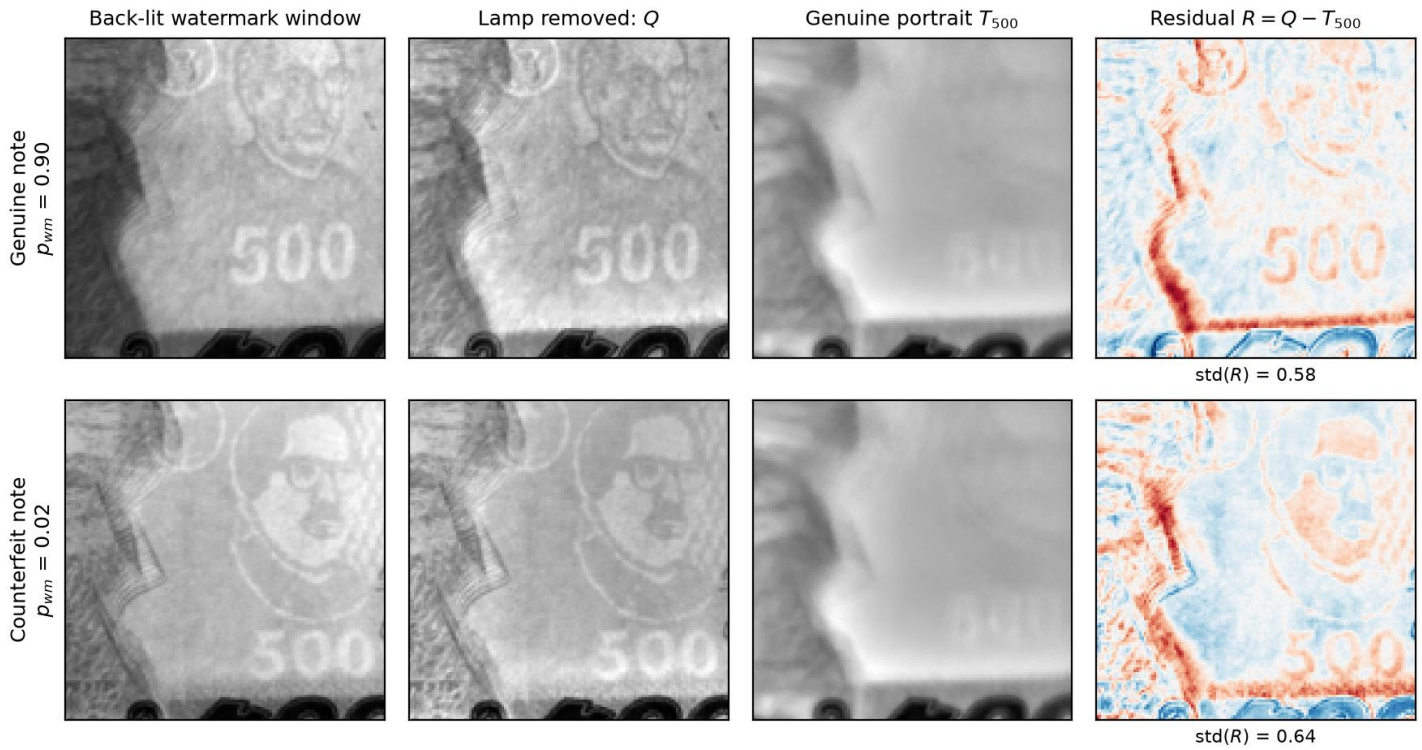
95.0%

on counterfeit prints never seen in training, with the watermark check

Why a stricter split: 279 of 322 readable 500 Taka counterfeits share one serial number, so the usual split tests copies of known prints.

Data: [14] JaalTaka; serial-number audit and print-disjoint split are ours. Earlier Taka counterfeit work: [16] SVM on security features; [17] CNNs; [19] UV flecks + RPN.

DMWR: a tiny, physics-based watermark checker



1. Remove the lamp: Q = I / (G∗I)
2. Genuine portrait: T = median of Q over genuine notes
3. Residual R = Q − T → two small CNN streams → "watermark clear?"

91.5%

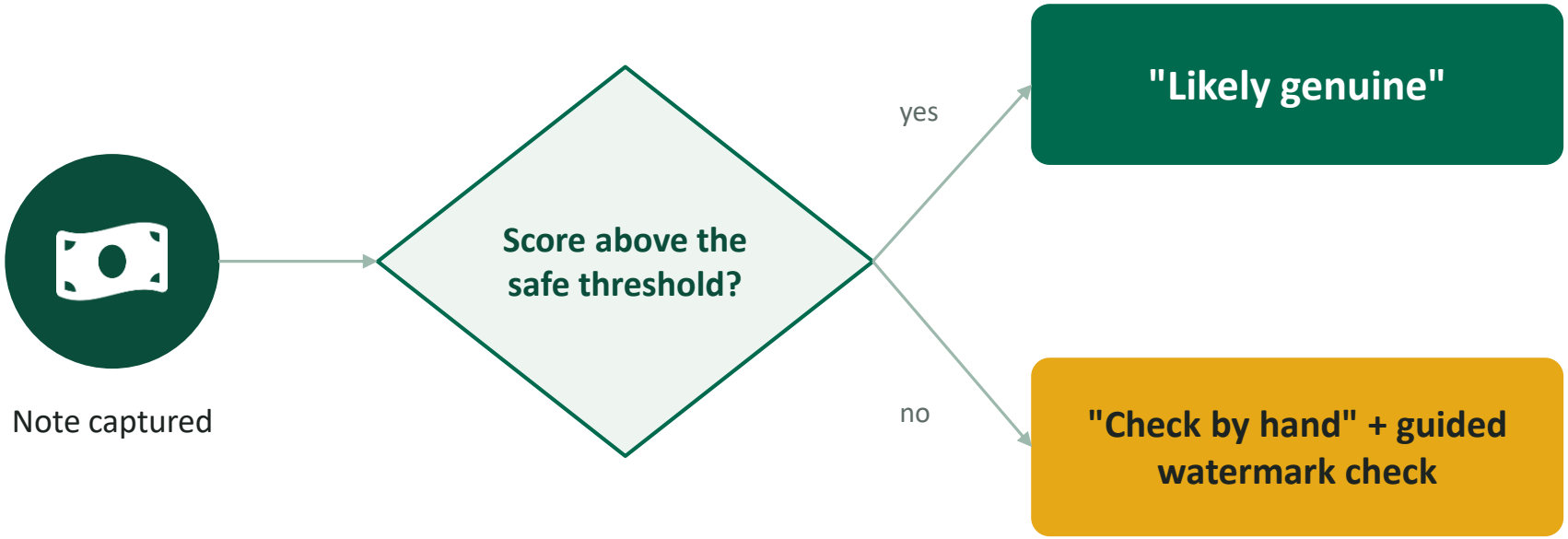
on unseen counterfeit prints (3 seeds, ±0.3)

46× smaller

than MobileNetV2 [18] (47,818 parameters, 0.21 MB), and as accurate: 92.9%, p ≥ 0.375

Baseline [18] Sandler et al., MobileNetV2, CVPR 2018; data [14] JaalTaka. DMWR (denomination-matched watermark residual) is our method.

Safe before decisive: the glass never says "counterfeit"



0 / 88

test counterfeits passed as "likely genuine"

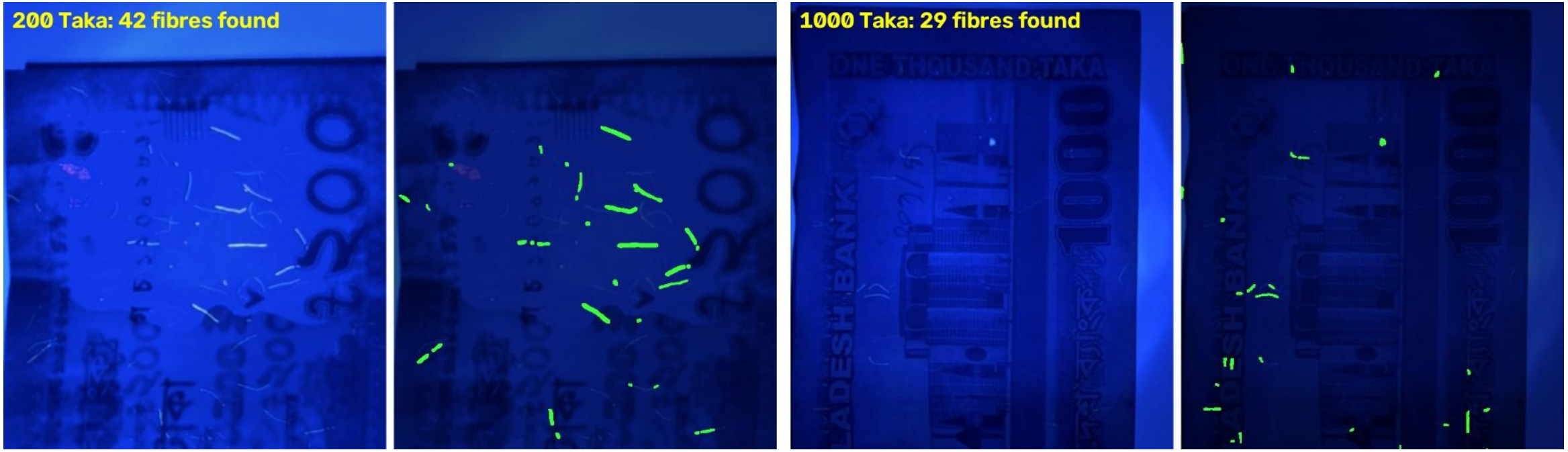
"Likely genuine": the score is higher than every counterfeit seen in validation (57.5% of genuine notes).

"Check by hand": the glass is not sure. It never means "counterfeit".

A screening aid, not a guaranteed counterfeit detector

Test notes: [14] JaalTaka.

UV fibre check: the glass carries its own UV lamp



85.6%

of 111 real UV photos of genuine notes accepted

88.6%

of SIMULATED fibre-free sheets rejected (not real counterfeits)

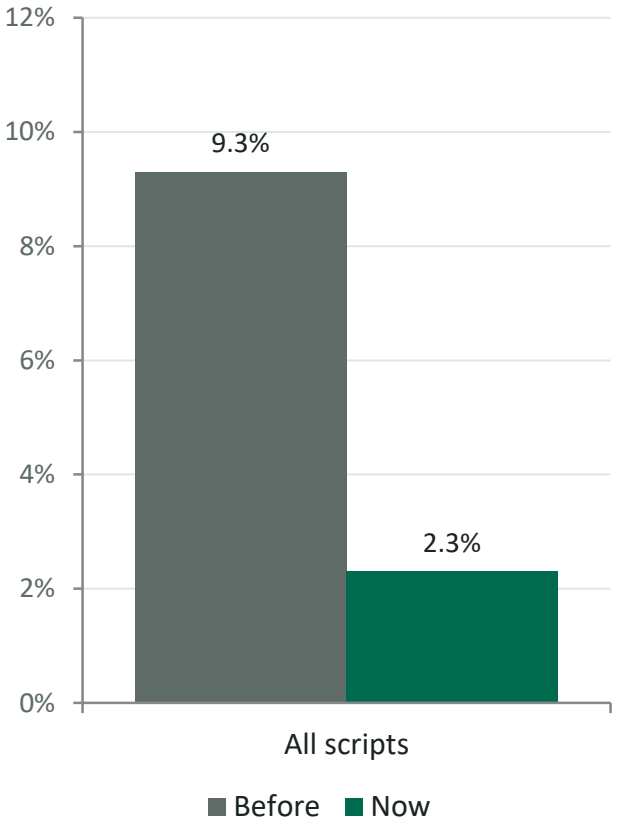
Prototype, off by default

The rejection result is simulated; it must be validated on real counterfeit notes

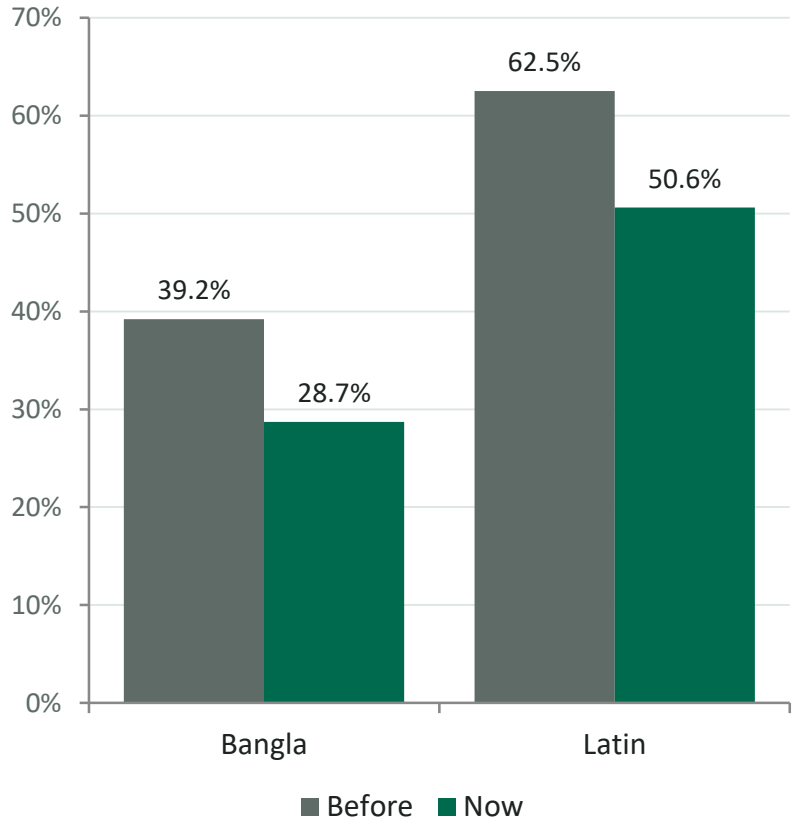
The glass's lamp emits 360-395 nm, covering the 365 nm used for banknote checks; banknote paper stays dark while its fibres glow [20]. Prior Taka UV work: [19] (96.3%). Fibre check and photos are ours.

Bangla reading: rendered text and real street photos

Rendered text (clean)



Real word photos (hard)



What helped

- Reading the main text block only
- Tuned detector and clean-up
- Recogniser fine-tuned on real words

What did not

Binarisation, Tesseract, TrOCR, PaddleOCR, dictionaries

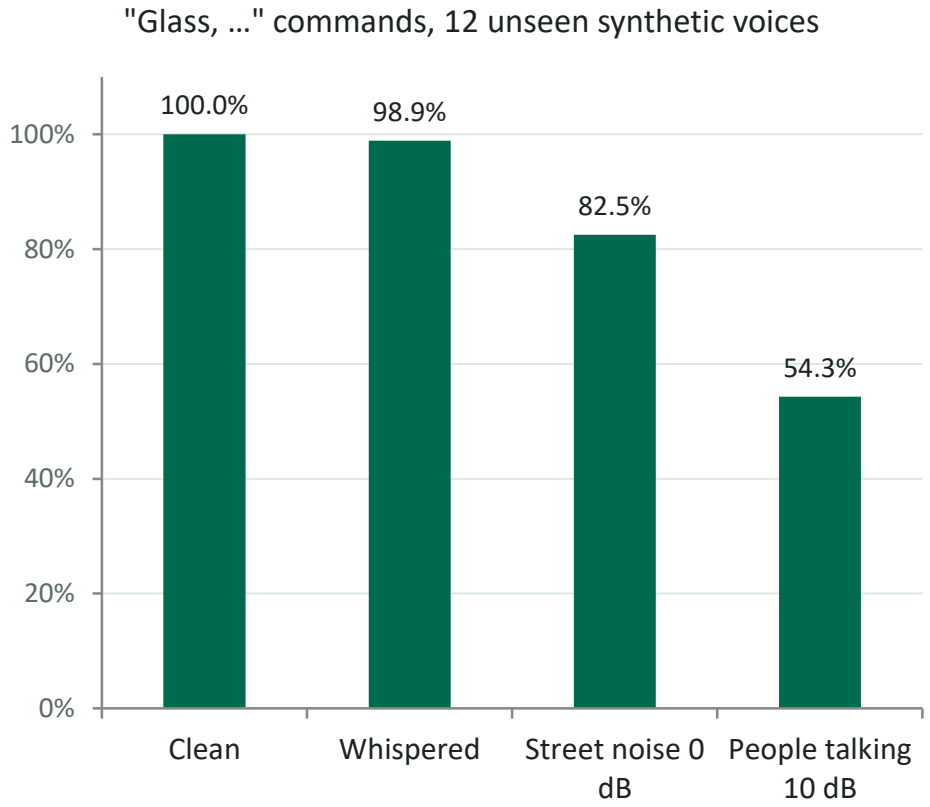
Character error rate, lower is better. Measured on evaluation datasets, not yet through the glass.

Engine [6] EasyOCR with [5] CRAFT; real words from [21] ICDAR 2019 MLT (1,000 test words). Every choice made on validation data.

Voice: a natural Bangla voice and spoken commands

Speaking

- Natural Piper voice, Bangla and English [10, 11]
- No broken sound when the Pi is busy
- "পড়ছি" (reading) while text is captured



212 → 13

false commands in 5,040 ordinary sentences, thanks to the wake word

0.86 s

from the end of speech to the action (median)

Command results: synthetic voices on a laptop, not human trials on the glasses

Recogniser [22] Vosk; noise from [23] Piczak, ESC-50; voices [10, 11]. Results: paper_evidence/VOICE_COMMANDS_FAST.md.

Running on the Raspberry Pi 5

399 ms
currency check, end to end

64 ms
watermark check

562 ms
object scan (YOLOv8s)

680 ms
emotion, face in view

18.3 s
3-line sign, text found once

1.48 GB
peak memory, 30-minute run



Key finding: heat

88.7 °C
peak temperature; the Pi throttled in 76% of samples (idle 71 °C)

Response: a thermal governor slows background scans when hot; better cooling is future work.

Measured on a Raspberry Pi 5 (8 GB) on 13 October 2026, board at its temperature limit; memory from the 1 October run. Changes after 13 October are not yet timed on the Pi.

Testing the worn glass: where the project stands



Prototype version 2 in a staged shop exchange by team members; illustration only, no error rate counted.

DONE — **Engineering checks**

Each part tested on held-out data, on a laptop and the Pi 5

PRELIMINARY — **First field test**

Glass camera: currency unreliable, Bangla text not read

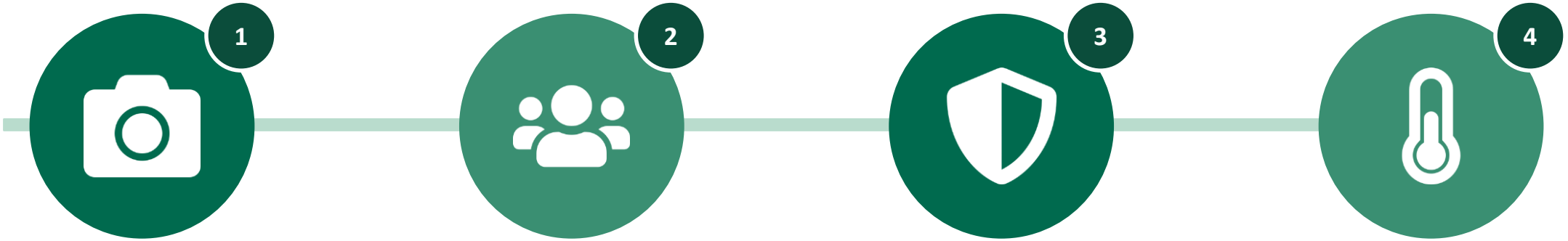
IN PROGRESS — **Participant trials**

Sessions with a visually impaired participant have started

NONE YET — **Completed user study**

No results about users are claimed

Future work: the roadmap



Camera

Higher resolution or autofocus

User study

8–12 participants, including their own voice commands

More banknotes

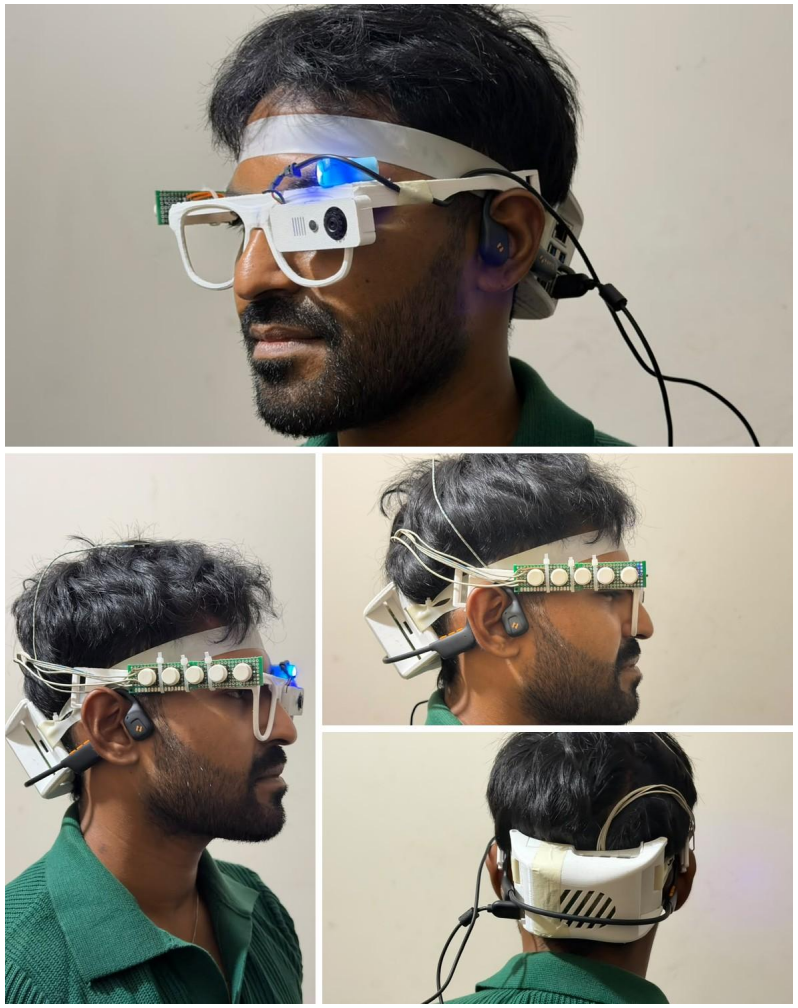
Unseen counterfeit prints; UV check on real fakes

Speed and cooling

Time the new code on the Pi; better cooling; battery

Done: engineering tests **In progress:** participant trials **Future:** the four steps above

Conclusion



Built

An offline Raspberry Pi 5 glass that reads Bangla and English, names objects and Taka notes, and speaks Bangla

Investigated

Synthetic training data, prefix training, unseen prints, a safe verdict, DMWR, hand-held notes, OCR, voice

Still to validate

Real counterfeits (also under UV), people's voices, the glass camera, a completed user study

Thank you

Questions?



References

[1] World Health Organization. Blindness and vision impairment: fact sheet, 2023.

[2] OrCam Technologies. OrCam MyEye.

[3] Ali A., H. et al. A Google Glass based real-time scene analysis for the visually impaired. IEEE Access 9, 2021.

[4] Brilli, D. D. et al. AIris: an AI-powered wearable assistive device for the visually impaired. arXiv:2405.07606, 2024.

[5] Baek, Y. et al. Character region awareness for text detection (CRAFT). CVPR, 2019.

[6] JaidedAI. EasyOCR: ready-to-use OCR with 80+ languages.

[7] Jocher, G., Chaurasia, A., Qiu, J. Ultralytics YOLOv8.

[8] Lin, T.-Y. et al. Microsoft COCO: common objects in context. ECCV, 2014.

[9] Li, S., Deng, W., Du, J. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild (RAF-DB). CVPR, 2017.

[10] Rhasspy. Piper: a fast, local neural text-to-speech system.

[11] eSpeak NG text-to-speech.

[12] A diverse image dataset for Bangladeshi currency recognition (BanglaTaka). Mendeley Data.

[13] Rafi, M. J. A., Rony, M., Majadi, N. NSTU-BDTAKA: an open dataset for Bangladeshi paper currency detection and recognition. Data in Brief 55, 2024.

[14] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Mendeley Data V2, 2025.

[15] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Data in Brief 64, 112377, 2026.

[16] Uddin, M. S., Das, P. P., Roney, M. S. A. Image-based approach for the detection of counterfeit banknotes of Bangladesh. ICIEV, 2016.

[17] Pathak, A. et al. Enhanced counterfeit detection of Bangladesh currency through convolutional neural networks. IJIRCST 12(2), 2024.

[18] Sandler, M. et al. MobileNetV2: inverted residuals and linear bottlenecks. CVPR, 2018.

[19] Chowdhury, A. A. et al. Evaluation of deep learning models on UV ink: a fake money detection scheme with RPN. ICCA (ACM), 2022.

[20] Mutanen, J., Jaaskelainen, T., Parkkinen, J. P. S. Luminescent security properties of banknotes. IS&T PICS Conference, 2003, pp. 421-424.

[21] Nayef, N. et al. ICDAR2019 robust reading challenge on multi-lingual scene text (RRC-MLT-2019). ICDAR, 2019.

[22] Alpha Cephei. Vosk offline speech recognition toolkit.

[23] Piczak, K. J. ESC: dataset for environmental sound classification. ACM Multimedia, 2015.

Links

Our code and results

SARA: glass software for the Raspberry Pi 5
github.com/aflam330/SARA

Research repository: experiments, results, thesis
github.com/aflam330/Senior_project_wearable_glass_for_blind_people_using_ocr_and_ylo

Datasets

JaalTaka (counterfeit notes)
data.mendeley.com/datasets/2m7wk5cy4c/2

BanglaTaka (note photos)
data.mendeley.com/datasets/3cv2sypkkh/1

NSTU-BDTAKA (hand-held notes)
www.sciencedirect.com/science/article/pii/S2352340924006681

Microsoft COCO
cocodataset.org

ICDAR 2019 MLT
rrc.cvc.uab.es/?ch=15

ESC-50 environmental sounds
github.com/karolpiczak/ESC-50

Open-source tools

Ultralytics YOLOv8
github.com/ultralytics/ultralytics

EasyOCR
github.com/JaidedAI/EasyOCR

Piper text-to-speech
github.com/rhasspy/piper

Vosk speech recognition
alphacephei.com/vosk/

eSpeak NG
github.com/espeak-ng/espeak-ng

WHO vision fact sheet
www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment